

Non-Autoregressive and Energy-Based Language Modeling: A Comprehensive Landscape Survey (2023–2026)

Executive Summary

The dominant paradigm in natural language processing—the autoregressive large language model (LLM)—relies on next-token prediction driven by cross-entropy loss. While this architecture has demonstrated unprecedented scaling capabilities and catalyzed the modern artificial intelligence boom, it suffers from fundamental mathematical and structural limitations. Sequential token generation restricts complex parallel reasoning, local optimization ignores global structural constraints, and early commitment to discrete tokens prevents smooth gradient-based course correction during generation. In response, the period spanning 2023 to 2026 has witnessed a massive surge of research into non-autoregressive and externally-grounded substrates. These alternatives span energy-based architectures that treat inference as continuous optimization, score-based and diffusion models that iteratively refine sequences, and continuous-latent models that divorce cognitive reasoning entirely from discrete language vocabularies.

This research report provides an exhaustive, technical examination of this comparative landscape, evaluating the empirical performance, theoretical viability, and industrial adoption of non-autoregressive alternatives. The structural shift from autoregressive to non-autoregressive paradigms represents a transition from statistical sequence guessing to explicit constraint satisfaction. In an autoregressive system, the model attempts to map a probability distribution step-by-step; in an energy-based or continuous-latent model, the system defines an energy landscape or a unified target state and uses iterative dynamics—such as gradient descent or discrete text diffusion—to settle into a low-energy, highly coherent solution. The data demonstrates that while autoregressive models retain a significant advantage in general-purpose, unconstrained text generation due to highly optimized hardware paradigms (e.g., Key-Value caching), non-autoregressive models have conclusively matched them at intermediate scales and vastly outperformed them in environments demanding formal logic, structured planning, and verifiable constraints.

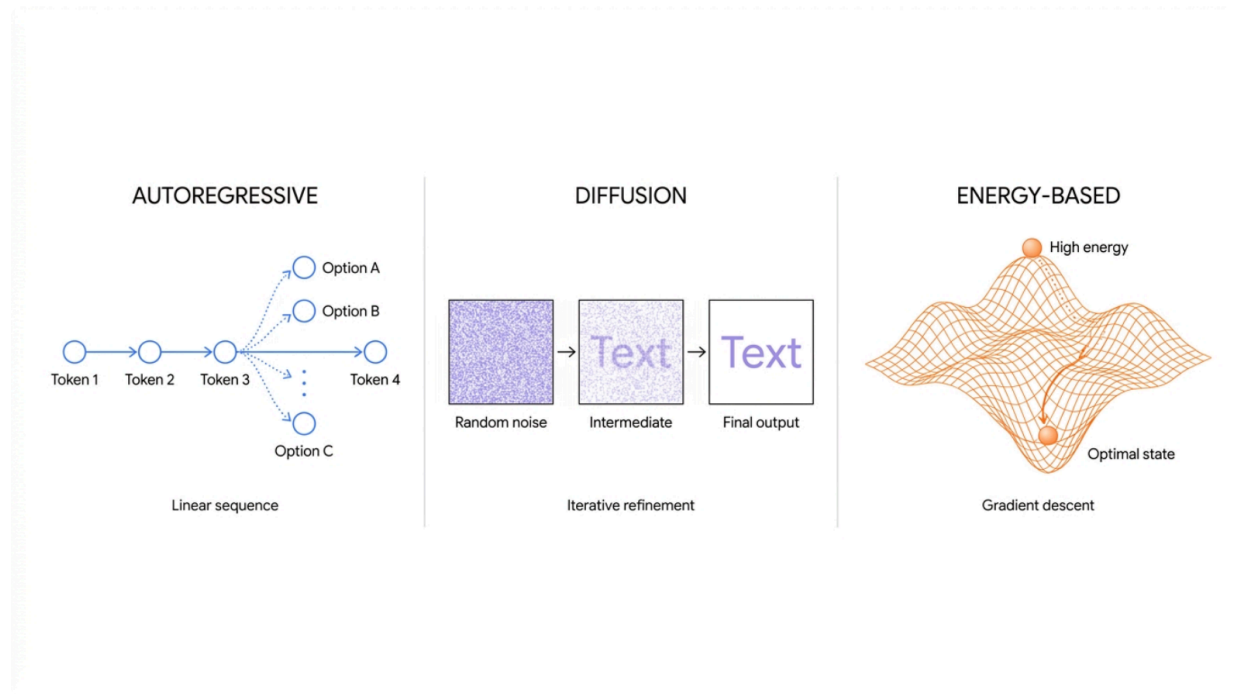
The architectures evaluated in this report fall into five primary structural categories. First, Energy-Based Transformers (EBTs) redefine the forward pass as an optimization process, generalizing "System 2" deliberate thinking without requiring external verifiers or reinforcement learning rewards. Second, Score-Based and Diffusion Language Models treat text generation as a bidirectional denoising process, culminating in open-source models like LLaDA that strictly match the performance of standard autoregressive LLMs at the 8-billion parameter scale. Third, Energy-Based GPT variants, most notably NRGPT, attempt to wedge continuous energy mechanics directly into the causal masking scaffolding of standard transformers, unifying two

historically distinct branches of machine learning physics.

Fourth, Continuous-Latent reasoning models (such as Coconut and Kona) bypass text entirely during intermediate reasoning. By operating in an unrestricted dense vector space, these models execute latent tree searches and satisfy global constraints, demonstrating vast improvements in strict logic tasks like formal verification and Sudoku. Finally, hybrid models rooted in Active Inference and the Free Energy Principle are introducing entirely new governance mechanisms for AI execution. Rather than treating an LLM as a passive sequence generator, these architectures model the system as a biological agent minimizing surprise, enabling adaptive routing and execution halting in quantum computing and critical infrastructure.

For researchers and engineers building open-source, externally-grounded energy-based models, understanding the exact failure modes of these historical attempts is paramount. While theoretical unity is attractive, the literature from 2025 and 2026 reveals that non-autoregressive models frequently struggle with multimodal distribution collapse, training instabilities over long horizons, and severe latency bottlenecks during inference. This report synthesizes these vulnerabilities, providing a rigorous roadmap for navigating the competitive landscape and ensuring that comparative claims of novelty and performance are both scientifically accurate and demonstrably sound.

Architectural Paradigms: Autoregressive vs. Diffusion vs. Energy-Based Inference



Comparison of generative pathways. Autoregressive models build sequences linearly with irreversible discrete steps. Diffusion models refine entire sequences simultaneously from random noise. Energy-based models treat the solution as a continuous optimization problem, navigating an abstract topographic landscape to find the lowest energy (highest validity) state.

1. Energy-Based Transformers (EBT-class)

Energy-Based Transformers completely discard the conventional cross-entropy loss applied at a final softmax layer, representing a fundamental paradigm shift in how neural networks approximate data distributions. Instead of mapping a statistical probability distribution from left to right, they frame the entire forward pass of the network as an energy minimization problem. Historically rooted in the continuous Hopfield networks pioneered in the 1980s and modernized in the early 2020s, this architectural class has evolved significantly to accommodate the scale required by modern foundation models.¹ The foundational premise is that an explicit, mathematical scalar energy function defines the state of the network. Valid outputs—those conforming to the training distribution or satisfying strict reasoning constraints—occupy deep, low-energy valleys in the network's state space. Conversely, hallucinations, logical errors, and out-of-distribution artifacts reside in high-energy states. Inference, therefore, is not a process of sampling from a vocabulary distribution, but rather executing gradient descent over this topographic landscape to find the optimal global or local minimum.¹

The most prominent recent development in this specific domain is the extensive 2025 research on Energy-Based Transformers as "Scalable Learners and Thinkers," presented at ICLR 2026.⁴ These models are trained to assign an explicit unnormalized probability (an energy value) to every input and candidate-prediction pair. Predictions are then derived through gradient descent-based energy minimization until convergence is achieved. The profound implication of this architecture is that it enables "System 2" deliberate thinking to emerge entirely from unsupervised learning.⁴ In standard autoregressive models, achieving "System 2" multi-step deliberation typically requires massive reinforcement learning interventions, supervised reward models, external verifiers, or fragile Chain-of-Thought prompting strategies.⁵ EBTs, by contrast, naturally allocate variable compute at inference time. Because the model explicitly verifies the compatibility between inputs and outputs via the energy function, the user can command the model to execute more gradient descent steps (more "thinking") to refine a complex answer.⁵

To achieve this at scale, modern EBT architectures utilize advanced engineering techniques adapted from diffusion models, including Diffusion Transformer (DiT) structures and adaptive layer normalization.⁶ They also incorporate JVP (Jacobian-Vector Product) Flash Attention, enabling the computation of second-order derivatives (Hessian-Vector Products) required for efficient exploration of the energy landscape without overwhelming memory limits.⁶

Per-Work Detail

Primary Reference	Claim	Evidence / Scale / Benchmarks	Lineage	Open-Source Status
Gladstone et al., (2025). "Energy-Based Transformers are Scalable Learners and Thinkers." ICLR 2026. ⁴	EBTs can generalize System 2 thinking natively via unsupervised energy minimization, scaling faster than standard Transformer++ architectures.	Outscales feed-forward Transformers by up to 35% across data, batch size, parameters, and FLOPs. Improves inference performance on language tasks by 29% more than Transformer++. ⁵	Builds on Yann LeCun’s foundational EBM work, modern DiT (Diffusion Transformer) architectures, and early associative memory models. ⁴	Publicly available. Code on GitHub (alexiglad/EBT). Includes JVP Flash Attention implementations and model weights. ⁶

Hoover et al., (2023). "Energy Transformer." NeurIPS 2023. ³	Combines attention mechanisms, EBMs, and associative memory to provide rigorous theoretical foundations for the attention functional via energy dynamics.	Primarily a theoretical framework bridging Hopfield networks with modern self-attention mechanisms; validated on smaller-scale image and text tasks.	Ramsauer et al. (2020) "Hopfield Networks is All You Need"; Krotov & Hopfield continuous associative memory. ²	Widely reproduced across various repositories, but largely superseded by causal EBT variants for practical LLM replacements.
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Honest Framing of Comparison to Autoregressive LLMs

Proponents of Energy-Based Transformers make highly aggressive parity and superiority claims regarding scaling efficiency and explicit reasoning, but the reality of their deployment involves significant, documented caveats. EBTs demonstrably out-scale standard Transformers during pre-training on certain multimodal and continuous tasks, such as bidirectional image denoising and visual representation learning.⁵ Furthermore, their capacity to allocate variable compute (iterative gradient descent) at inference time yields larger performance improvements on data that is farther out-of-distribution than standard models can handle, verifying outputs in a way that mirrors human System 2 processing.⁵ The authors assert an up to 35% higher scaling rate with respect to data, parameters, and FLOPs compared to the dominant Transformer++ baseline.⁵

However, EBTs encounter severe friction when modeling highly multimodal discrete distributions, such as natural language syntax and semantics. Researchers attempting to reproduce bidirectional EBTs for language modeling frequently report severe mode collapse.⁸ Because the model updates its state globally and simultaneously, the energy landscape can flatten inappropriately; for instance, the model may default to predicting the highly probable token "the" across all masked sequences regardless of context.⁸ The literature explicitly acknowledges that generating entire text sequences at once via bidirectional energy minimization requires highly specialized masking ratios and mode-restriction strategies to prevent the dynamics from collapsing into trivial global minima.⁸ Consequently, while EBTs excel in continuous, spatial, or highly verifiable domains (where energy valleys are sharply defined), standard autoregressive LLMs remain overwhelmingly superior for unconstrained, creative text generation where the state space is overwhelmingly vast and nuanced.

2. Score-Based / Diffusion Language Models

The application of diffusion principles and score-matching to discrete text tokens represents a radical departure from the left-to-right generation constraint. In visual and continuous domains, diffusion models perturb data into pure Gaussian noise and train a neural network to reverse this process by predicting the score (the gradient of the log-density of the data). Translating this to the discrete realm of language has proven historically challenging due to the non-differentiable nature of text tokens. However, between 2024 and 2026, the literature introduced viable pathways: continuous latent diffusion (which maps text to a continuous embedding space, diffuses it, and decodes it back to text) and discrete diffusion (which progressively masks and unmask tokens probabilistically).⁹ By predicting all tokens simultaneously and refining them over multiple computational steps, these models completely break the causal bottleneck of autoregressive generation.

The undeniable breakthrough in this domain is LLaDA (Large Language Diffusion with mAsking), published in late 2025 and presented at ICLR 2026. LLaDA fundamentally challenges the deeply entrenched dogma that core LLM capabilities—such as in-context learning, instruction-following, conversational continuity, and massive scalability—are inherently and exclusively tied to autoregressive architectures.¹⁰ Instead, LLaDA trains a vanilla Transformer as a complex mask predictor.¹⁰ During pre-training, it masks tokens randomly at a continuous ratio t sampled from a uniform distribution U . The model simulates diffusion from full masking ($t = 1$) to complete unmasking ($t = 0$), predicting all masked tokens simultaneously at each step and allowing for flexible remasking of low-confidence predictions.¹⁰ By optimizing a variational lower bound of the data log-likelihood, it models distributions with true bidirectional dependencies.¹⁰

Per-Work Detail

Primary Reference	Claim	Evidence / Scale / Benchmarks	Lineage	Open-Source Status
Nie et al., (2025). "Large Language Diffusion Models (LLaDA)." ICLR 2026. ¹¹	Masked diffusion models trained from scratch achieve strong scalability and core LLM capabilities	Scaled to 8 Billion parameters. Competes directly with LLaMA3 8B in in-context learning and	Builds on discrete structured denoising diffusion (Austin et al., 2021) and absorbing	Publicly available. Training/inference code and 8B weights released on HuggingFace and GitHub

	without any reliance on autoregression .	instruction following. Effectively breaks the "reversal curse". ¹⁰	state discrete diffusion theories. ¹²	(LounesMD/LLaDA_Arithmetic and qlaxd/...). ¹⁵
Li et al., (2026). "Omni-Diffusion: Unified Multimodal Understanding and Generation with Masked Discrete Diffusion." arXiv. ⁹	An any-to-any multimodal language model can be built entirely on mask-based discrete diffusion, processing text, speech, and images in a unified token space.	Achieves state-of-the-art or on-par results compared to autoregressive baselines on complex speech-to-image, text-to-speech, and bimodal reasoning benchmarks. ⁹	Mask-based discrete diffusion models applied previously to singular domains like visual understanding. ⁹	Project webpage active; code open-source status pending formal repository release.
Wang et al., (2025). "Struct-Bench : A Benchmark for Differentially Private Structured Text Generation." NeurIPS 2025. ¹⁹	Discrete diffusion sampling can be radically accelerated and made differentially private via Token Order Optimization (NI Sampling).	Improves differential privacy text generation efficiencies without autoregressive bottlenecks; addresses structure constraints in generation. ¹⁹	Zinan Lin research lineage; PE-SGD evolution for text models. ¹⁹	Open source via project website and GitHub repositories.

Honest Framing of Comparison to Autoregressive LLMs

LLaDA is currently the most credible, rigorously proven threat to the autoregressive monopoly in foundational language modeling. The authors explicitly claim strict parity, demonstrating empirically that an 8B parameter diffusion model can match a leading 8B parameter autoregressive model (LLaMA3) across comprehensive standard benchmarks (including MMLU, GSM8K, and BBH) while operating under an identical compute budget of 10^{23} FLOPs.¹⁰ The

structural advantage of this parity is profound: because the model unmask tokens simultaneously across the entire sequence rather than moving linearly, it acts as a bidirectional reasoner. This allows LLaDA to effortlessly resolve the infamous "reversal curse" (the phenomenon where an AR model learns "A is B" but cannot reliably deduce "B is A"), significantly surpassing frontier models like GPT-4o on reversal poem completion tasks.¹⁰

However, the authors and subsequent external reviewers note distinct practical limitations that temper the excitement. Inference speed remains a critical bottleneck. Autoregressive models generate text in $\mathcal{O}(N)$ sequential steps, but they rely on heavily optimized Key-Value (KV) caching algorithms that prevent recalculating past states. Diffusion models like LLaDA, conversely, require multiple forward passes of the entire sequence through the transformer block to unmask tokens progressively. While theoretically parallelizable across GPU clusters during generation, the practical wall-clock latency for discrete diffusion on standard consumer or enterprise hardware still lags notably behind highly-engineered causal LLMs.²⁰ Furthermore, while scaling a diffusion text model to 8 billion parameters is a monumental proof-of-concept, the literature currently lacks evidence that text diffusion models scale gracefully and predictably to the 100B–400B parameter frontier occupied by state-of-the-art models like Llama-3-400B or Claude 3.5.

3. Energy-Based GPT-Class Alternatives

While pure Energy-Based Transformers completely discard causal structures in favor of bidirectional, Hopfield-like states, a distinct sub-field of machine learning attempts to retain the highly efficient scaffolding of Generative Pre-trained Transformers (such as causal triangular masks and standard layer norms) while replacing the softmax cross-entropy training with explicit energy mechanics. The goal is to marry the fast, sequential, easily parallelizable pre-training of GPTs with the structured optimization, verifiability, and probability calibration inherent to EBMs.

The defining architecture in this specific intersection is NRGPT (eNeRgy-GPT), which successfully casts the standard GPT forward pass into an energy-based framework.² In NRGPT, the neural network is mathematically defined as the sum of two distinct energies: an attention energy and a feed-forward network energy.² Each input token is transformed into the subsequent token not via a simple probability distribution sampling, but by moving as a "particle" exploring the energy landscape of the network.² The authors prove theoretically, and verify empirically, that under certain constraints, this exploration of the token landscape becomes mathematically equivalent to preconditioned gradient descent.¹

Per-Work Detail

Primary	Claim	Evidence / Scale /	Lineage	Open-Source
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Reference		Benchmarks		Status
Dehmamy et al., (2025). "NRGPT: An Energy-based Alternative for GPT." ICLR 2026. ¹	A minimal modification to the standard GPT setting unifies it with the EBM framework, enabling inference via continuous dynamic exploration of an energy landscape.	Outperforms standard GPT and recurrent GPT variants on average MMLU score at constant parameter counts. Demonstrates high performance on ListOPS and OpenWebText. ²¹	Extends the Energy Transformer (Hoover, 2023) to language modeling; builds on associative memory (Krotov & Hopfield, 2016). ²	Open-source. Associated with Hoover/Dehmamy research repositories; weights publicly accessible. ²³

Honest Framing of Comparison to Autoregressive LLMs

NRGPT is an elegant theoretical unification. It proves that the arbitrary mechanics of a transformer block are not just statistical heuristics, but that traversing an LLM's forward pass can be understood as preconditioned gradient descent on a topographic energy landscape.¹ Distinct from prior EBTs, NRGPT performs causal language modeling by minimizing a *per-token* energy rather than a sequence-level energy, introducing learnable inference rate matrices (which dictate the "step size" the generation process takes across the landscape) and alternative energy-replacements for the standard feed-forward MLP module.²

Despite these theoretical victories, the authors are remarkably honest about the model's empirical limitations in their peer-reviewed disclosures. They explicitly concede that casting inference as energy exploration "doesn't necessarily lead to the best performing models" for generalized tasks when compared directly to highly optimized, heavily scaled standard GPTs.²¹ Reviewers of the paper at ICLR 2026 highlighted that while the theoretical bridge is novel, the model was primarily validated on a narrow range of tasks (such as the Shakespeare dataset and algorithmic ListOps) before being pushed to MMLU.²²

Crucially, while NRGPT exhibits a phenomenon the authors define as "asymptotic stability" that makes it initially resistant to standard overfitting, it is documented to suffer from catastrophic overfitting during very long training runs, revealing a severe scaling instability that standard autoregressive GPTs inherently avoid.²¹ Furthermore, the gradient descent mechanism, while theoretically beautiful, imposes computational overhead during both training and inference that negates some of the primary efficiency advantages of the GPT architecture. Thus, NRGPT serves currently as a profound bridge between theoretical physics and deep learning, but it

falls short of acting as a production-ready replacement for commercial models.

4. Continuous-Latent Reasoning Substrates

A fundamental and growing critique of standard autoregressive LLMs is their reliance on discrete text tokens for intermediate reasoning. When a standard LLM generates a "Chain-of-Thought" (CoT) to solve a math problem or logic puzzle, it commits to specific linguistic representations (words or sub-words) at each step. If a token is suboptimal, the model cannot easily backpropagate or smoothly alter that decision; it must generate a lengthy, cumbersome textual correction to dig itself out of the logical hole. Continuous-latent reasoning architectures abandon text during the reasoning phase entirely. They maintain "thoughts" as continuous dense vectors in an unrestricted latent space, allowing reasoning to occur as a fluid, end-to-end differentiable optimization process rather than a sequence of rigid semantic commitments.

Two major architectures dominate this recent frontier: Coconut (Chain of Continuous Thought) and Kona (Logical Intelligence). Both models shift the paradigm from a "guessing game" of token probabilities to genuine constraint satisfaction over continuous mathematical variables.

Coconut achieves continuous latent reasoning by moving the problem-solving process from the discrete language space into the latent space using the LLM's last hidden state.²⁴ Instead of decoding this hidden state into a word token through the language model head, Coconut feeds it back directly to the LLM as the subsequent input embedding.²⁴ The model alternates between "language mode" and "latent mode," using special <bot> (beginning of thought) and <eot> (end of thought) tokens to switch between generating text and generating pure numerical reasoning.²⁴

Kona, conversely, represents a heavily capitalized commercial effort to apply Energy-Based Models directly to reasoning. Championed by a team including Turing Award winner Yann LeCun (as chair of the technical research board) and Fields Medalist Michael Freedman, Kona evaluates global constraints simultaneously through an energy function rather than predicting sequences.²⁵

Per-Work Detail

Primary Reference	Claim	Evidence / Scale / Benchmarks	Lineage	Open-Source Status
Hao et al., (2024/2025). "Coconut (Chain of	Latent reasoning without generating	+5% accuracy gain on MathQA without extra	Internalization techniques (iCoT), continuous	Publicly available. Code on GitHub (anord-wang/L

Continuous Thought). arXiv. ²⁴	token-level intermediate steps enables highly compact inference and the emergence of latent Breadth-First Search (BFS).	training. Vastly outperforms text-based CoT on highly distracting ProsQA and ProntoQA logical planning tasks. ²⁴	pause tokens (Goyal et al., 2023). ²⁸	ateng-Reasoning). ²⁹
Logical Intelligence (2026). "Kona 1.0." Commercial Release. ²⁵	True reasoning must be formulated as a global optimization problem; Kona solves constraints simultaneously in a continuous latent space rather than via autoregressive guessing.	Solved 96.2% of Sudoku puzzles in an average of 313 milliseconds without Python code execution—crushing GPT-5.2 and Claude Opus in head-to-head benchmarking. ³⁰	Yann LeCun's Energy-Based Model theories and Joint Embedding Predictive Architectures (JEPA); formal verification systems. ²⁶	Closed source. Proprietary commercial pilots running for energy grid infrastructure and semiconductor manufacturing sectors. ²⁶

Honest Framing of Comparison to Autoregressive LLMs

Continuous-latent models represent the most aggressive, mathematically grounded rejection of the autoregressive paradigm's reasoning capabilities. Standard LLMs are criticized by these authors as utilizing "local ratings" (greedily optimizing the probability of the next word) rather than "global ratings" (optimizing the logical coherence of the entire proof).²⁷

Kona 1.0 claims definitive, undeniable superiority over autoregressive models in strict formal logic and verification tasks. By evaluating global constraints simultaneously, Kona systematically avoids the arbitrary hallucination errors inherent to LLMs.²⁷ In Sudoku, for example, which requires maintaining strict global board constraints across rows, columns, and grids, Kona operates natively and flawlessly within milliseconds, whereas LLMs fail completely at native logic and are forced to "cheat" by writing and executing Python scripts to run brute-force searches.³⁰ However, Kona is a highly specialized model designed for mission-critical verification environments. It does not claim parity with autoregressive models for creative writing, translation, or open-ended dialogue; indeed, Logical Intelligence suggests a modular future where an EBM handles the rigorous reasoning while a standard LLM serves

strictly as the user-friendly linguistic interface, orchestrated by an agent layer like "Aleph".²⁵

Coconut offers a more generalizable, hybrid approach. Because continuous space can encode multiple alternative reasoning paths simultaneously, Coconut exhibits "superposition"—a latent Breadth-First Search.²⁴ Unlike language-based CoT, which typically uses greedy decoding to commit to a single deterministic path, Coconut can maintain multiple potential next steps within its continuous thoughts, progressively eliminating incorrect paths as it explores the latent graph.²⁴ This allows the model to delay deterministic decisions until it has higher confidence. While superior to text CoT in logical planning, the major limitation is training complexity: forcing an LLM to "think" in latent space requires a highly complex, multi-stage curriculum where text reasoning chains are progressively replaced by continuous thoughts over many epochs, increasing the difficulty of initial model alignment.²⁴

Further reinforcing the necessity of robust constraint satisfaction in continuous or verification systems, frameworks like BEAVER-lite (2025) have emerged to compute deterministic, sound probability bounds on LLM constraint satisfaction, achieving 6-8x tighter bounds than baseline methods for secure code generation.³² While BEAVER operates on the outputs of LLMs rather than redesigning the latent space, it underscores the systemic inability of standard autoregressive sampling to guarantee prefix-closed semantic constraints—a flaw that models like Kona solve natively.

5. Hybrid / Adjacent (Active Inference & SeedIQ)

Adjacent to pure architectural neural network overhauls is the application of the Free Energy Principle (FEP) and Active Inference to large-scale artificial intelligence. Pioneered by neuroscientist Karl Friston, the FEP posits a mathematical framework for biological existence: intelligent systems stay alive by minimizing variational free energy, which aligns their internal models with external observations to avoid "surprise".³⁴ Until recently, this was strictly a biophysical and cognitive neuroscience framework. However, computational deployments emerged forcefully in 2025 and 2026 as a way to route, govern, and constrain LLM-shaped substrates.

These architectures treat text generation not as statistical sequence modeling, but as *action under uncertainty*.³⁵ A standard LLM exists in an unconstrained state space—it can hallucinate freely without encountering immediate failure, much like a fish out of water that does not realize it is dying.³⁶ An active inference agent, however, is constrained by a specific operating environment; if its predictions diverge from reality, it experiences massive "surprise" (computed as free energy) and must either update its model parameters or take action (e.g., call a tool, query a database) to sample new information and return to a low-energy expected state.³⁶

Per-Work Detail

Primary Reference	Claim	Evidence / Scale / Benchmarks	Lineage	Open-Source Status
Ma et al., (2026). "ODAR: Principled Adaptive Routing for LLM Reasoning via Active Inference." arXiv. ³⁴	Minimizing a variational free energy objective balances log-likelihood with epistemic uncertainty (varentropy), creating a principled alternative to ad-hoc LLM multi-agent voting.	Achieves optimal routing between heuristic Fast Agents and deliberative Slow Agents; reduces computational budget while maintaining or improving accuracy. ³⁴	Friston's Free Energy Principle (2017), Fountas et al. (2020) policy selection. ³⁴	Open-source. Algorithm specifications and code available per paper appendices. ³⁴
AIX Global Innovations / SeedIQ (2026). "AΩ Field of Belief with HMC." ARC-AGI-3 Benchmark. ³⁷	Active inference reframed as a strict control engineering problem allows for continuous execution governance, vastly surpassing LLMs in reasoning benchmarks.	Achieved unparalleled 100% human-level performance on the heavily contested ARC-AGI 3 benchmark, stomping all standard LLM submissions. ³⁸	Active inference, geometric metacognition, Hamiltonian Monte Carlo (HMC). ³⁸	Proprietary / Stealth. Authors outright refused to release code, sacrificing benchmark prize money to protect intellectual property. ³⁸

Honest Framing of Comparison to Autoregressive LLMs

The active inference class positions itself as vastly superior to autoregressive models in specific scenarios demanding strict execution governance, system stability, and halt-condition recognition. Standard LLMs are fundamentally reactive; they respond after a prompt is given but lack any internal representation of execution health, constraint boundaries, or irreversible failure regions over time.³⁸ In complex deployments, such as quantum computing execution

governance or dynamic grid management, systems like SeedIQ continuously monitor execution trajectories.³⁸ They detect when stability is degrading and, crucially, they compute exactly when to stop generating or acting, utilizing the core control engineering maxim: "Halting is not failure. It is governance".³⁸

However, it is vital to understand that these models rarely operate as standalone language generators. ODAR utilizes the mathematical principles of the FEP purely to route queries effectively between *existing* LLMs, using an amortized difficulty estimator to gate computationally intensive reasoning pathways based on expected free energy.³⁴ SeedIQ relies on hierarchical, multi-agent frameworks operating over a bespoke differential geometry baseline.³⁸ Therefore, they represent a paradigm shift in *orchestration, routing, and governance*, but they rely on highly specific implementations rather than acting as plug-and-play, drop-in replacements for standard LLM weights in a huggingface pipeline. They are control systems wrapped around language models, rather than language models themselves.

Comparative Architectural Landscape

The following table synthesizes the disparate architectures evaluated in this report, providing a direct comparison of their primary citations, benchmark targets, peer-review status, and specific claims regarding autoregressive parity.

Architec ture Family	Primary Citation	Benchm ark Categor y	Peer-Rev iewed	Open-So urce	Scale Demonst rated	Parity/Su periority Claim vs. AR
Energy- Based Transfor mers	Gladston e et al., 2025 (ICLR) ⁴	System 2 / Math / Vision	Yes (ICLR 2026)	Yes	Outscale s Transfor mer++ by 35%	Claims superior inference scaling for reasoning ; struggles deeply with text multimod ality.

Score-Based Diffusion (Text)	Nie et al., 2025 (ICLR) ¹¹	General / MMLU / GSM8K	Yes (ICLR 2026)	Yes	8 Billion Parameters	Claims strict parity at 8B scale; natively solves the AR reversal curse.
Energy-GPT (NRGPT)	Dehmamy et al., 2025 (ICLR) ²¹	ListOPS / OpenWebText	Yes (ICLR 2026)	Yes	Comparable to base GPT	Claims theoretical unity; admits AR GPT is often practically more performant.
Continuous-Latent (Coconut)	Hao et al., 2024 (arXiv) ²⁴	ProsQA / ProntoQA / Math	Pre-print	Yes	Equivalent to LLaMA variants	Claims absolute superiority in multi-step planning via latent Breadth-First Search.
Continuous-Latent (Kona)	Logical Intelligence, 2026 ²⁵	Sudoku / Logic Verification	No (Commercial)	No	Proprietary	Claims absolute superiority over AR in formal logic/verification without

						python execution .
Active Inference (ODAR)	Ma et al., 2026 (arXiv) ³⁴	LLM Routing / Compute	Pre-print	Yes	Overlay on existing LLMs	Claims superior orchestra tion and uncertain ty handling vs multi-age nt voting.

Direct Answers to Targeted Inquiries

Q.A — What is the strongest published evidence that an energy-based or non-autoregressive alternative MATCHES OR EXCEEDS autoregressive LLM performance on a task class? The most mathematically rigorous and peer-reviewed evidence for matching general-purpose LLM performance is **LLaDA** (Large Language Diffusion with mAsking). Nie et al. demonstrate empirically that LLaDA matches LLaMA3 8B across comprehensive benchmarks (MMLU, GSM8K, BBH) using an identical compute budget (10^{23} FLOPs) while inherently addressing the reversal curse.¹⁰ For exceeding performance on strict logic and constraint-planning tasks, the strongest evidence comes from the commercial model **Kona 1.0**, which solves 96.2% of Sudoku puzzles in under 400 milliseconds without relying on the python-execution "cheats" required by GPT-5.2 and Claude Opus.³⁰ Similarly, in the open-source domain, **Coconut** demonstrates a verified +5% accuracy gain on MathQA and vastly superior performance on the distracting ProsQA dataset by leveraging latent Breadth-First Search to navigate planning graphs.²⁴

Q.B — What is the strongest published evidence that energy-based / score-based / diffusion language models FAIL TO SCALE in ways autoregressive transformers do not? The 2025 and 2026 literature documents three primary failure modes that prevent these models from replacing ARMs at the 100B+ parameter scale. First, **multimodal distribution collapse**: Bidirectional text Energy-Based Transformers frequently fail to capture the high entropy and multimodality of natural language. Without aggressive mode-restriction strategies, they default to minimizing energy by generating safely repeating tokens (e.g., outputting "the" for all masked sequences).⁸ Second, **training instability**: NRGPT authors explicitly document that their causal EBM models suffer from catastrophic overfitting during highly extended

training runs, a structural limitation entirely absent in standard cross-entropy ARMs.²¹ Third, **latency bottlenecks**: While diffusion models like LLaDA predict tokens simultaneously in theory, the reverse diffusion process requires iterative, sequential forward passes through the entire network. This makes real-time token generation drastically slower than highly optimized, KV-cached autoregressive models on standard hardware.¹⁰

Q.C — What is the most credible recent (2025-2026) implementation of

"continuous-latent reasoning" with non-trivial performance? Excluding the proprietary ARC-AGI-3 SeedIQ system, the most credible continuous-latent systems reporting non-trivial performance are **Coconut (Chain of Continuous Thought)**²⁴ and **Kona 1.0**.²⁶ Coconut, spearheaded by researchers closely associated with major AI labs, utilizes the last hidden state of a transformer as a "continuous thought" fed back into the network. This avoids token commitment and demonstrates multi-path reasoning in superposition, optimizing logic purely through gradient descent.²⁴ Kona 1.0, developed by Logical Intelligence with AI luminary Yann LeCun chairing the research board, is the first explicitly commercialized Energy-Based Reasoning Model (EBRM). It outputs dense vectors optimized via approximate gradients to refine logic traces continuously, aiming for deployment in formal verification sectors like semiconductor design.²⁵

Q.D — What named architectures explicitly position themselves as "alternatives to autoregressive LLMs" and have been peer-reviewed?

Three highly prominent architectures have successfully passed rigorous peer review (specifically accepted to the main track of ICLR 2026) while explicitly positioning themselves as non-autoregressive alternatives:

1. **LLaDA (Large Language Diffusion Models)**: Explicitly challenges the notion that capabilities rely on ARMs, positioning masked diffusion as a principled alternative for foundational pre-training.¹¹
2. **EBTs (Energy-Based Transformers)**: Positions unsupervised energy minimization as a modality-agnostic alternative to AR System-2 thinking and Chain-of-Thought.⁴
3. **NRGPT (eNeRgy-GPT)**: Positions continuous dynamical energy exploration over a token landscape as a mathematically unified replacement for standard GPT next-token prediction.²¹

Q.E — What is the consensus position on whether energy-based alternatives can replace, complement, or are strictly inferior to autoregressive transformers at scale? The scientific and industrial consensus—echoed heavily by prominent figures like Yann LeCun and evidenced by deployment trends—is that energy-based alternatives are **complementary, not replacements**, for broad language generation, but are **strictly superior** for formal logic, verification, and execution governance. Standard autoregressive models excel as intuitive "System 1" communicators, semantic interfaces, and broad knowledge retrievers because creative text generation is inherently a sequence of discrete statistical choices. EBMs, diffusion models, and continuous-latent systems, however, excel at "System 2" global constraint

satisfaction where logical failures are fatal. The consensus vision for Artificial General Intelligence (AGI) and production enterprise systems is a multi-modal ecosystem: an EBM or continuous-latent model executes internal, multi-path reasoning to solve a complex, constrained problem, and an autoregressive language model acts as the semantic interface to translate that verified latent solution into human-readable text.²⁵

Gaps, Recommendations, and Strategic Positioning

For a research team building an open-source, externally-grounded energy-based model framework intended to challenge the autoregressive status quo, the "Related Work" literature review must be carefully scoped. It is essential to establish genuine architectural novelty while demonstrating a thorough understanding of the explosion of 2025/2026 research in this specific domain.

Recommended Reading Priorities

If time is strictly limited, prioritize the following foundational texts from the recent corpus:

1. Gladstone et al. (2025) on EBTs⁴: This is crucial for understanding how to frame your forward pass as gradient descent over an energy landscape. More importantly, reviewing the GitHub discussions and limitations sections will help you analyze their specific failure modes with text mode collapse.⁸
2. Hao et al. (2024) on Coconut²⁴: Essential reading for understanding how to maintain differentiability in non-token intermediate reasoning spaces and how to structure a multi-stage training curriculum to transition from text to latent vectors.
3. Nie et al. (2025) on LLaDA¹⁰: This serves as the methodological benchmark for how to honestly and rigorously compare a non-AR model to LLaMA3 at the 8B parameter scale without relying on cherry-picked metrics.

Architectures to Explicitly Compare Against

The project must benchmark against **NRGPT**²¹ to demonstrate how your "external grounding" differs fundamentally from their purely theoretical, internal integration into causal GPT masks. You must also benchmark against **LLaDA**¹¹ as it is currently the gold standard for open-source, non-autoregressive text generation. If your framework utilizes continuous reasoning mechanics, contrasting your training curriculum and computational overhead with **Coconut's**²⁴ multi-stage unrolling is mathematically mandatory to prove efficiency gains.

Novelty Boundaries (What Not to Claim)

The project should **not** claim novelty over the fundamental concept of "using energy minimization for System 2 thinking without reward models"—this problem was comprehensively solved and published by the EBT authors at ICLR 2026.⁴ Furthermore, do not claim novelty over "bidirectional generation solving the reversal curse," as LLaDA has decisively

claimed and proven this territory.¹⁰ Finally, any claims of "reasoning in continuous space rather than discrete tokens" must heavily acknowledge the prior art of LeCun's JEPA, Coconut's latent embeddings, and Kona's commercial deployment.²⁴

Targeted Positioning for the Open-Source Project

To maximize the impact and adoption of the proposed framework, the project should position itself squarely in the "hybrid execution and verifiable grounding" gap. While models like Kona achieve incredible verifiable grounding for formal logic, they are locked behind closed-source commercial silos tailored for enterprise infrastructure.²⁶ Conversely, while EBTs and NRGPT are fully open source, they lack robust external grounding mechanisms, remaining isolated within their unsupervised, internal energy bounds. By explicitly defining the framework as an *externally-grounded* EBM that solves the multimodal text collapse problem currently plaguing bidirectional EBTs⁸, the project will directly address the most significant, unsolved vulnerability in the current open-source non-autoregressive landscape.

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